ASSIGNMENT NO. 1: Introduction to Databases and SQL Environment

Questions:

Q1) Write SQL commands to create a database named SchoolDB. The program should:

1. Create a new database named SchoolDB.

2. Select the database for use.

3. Display confirmation by creating a simple table inside it.

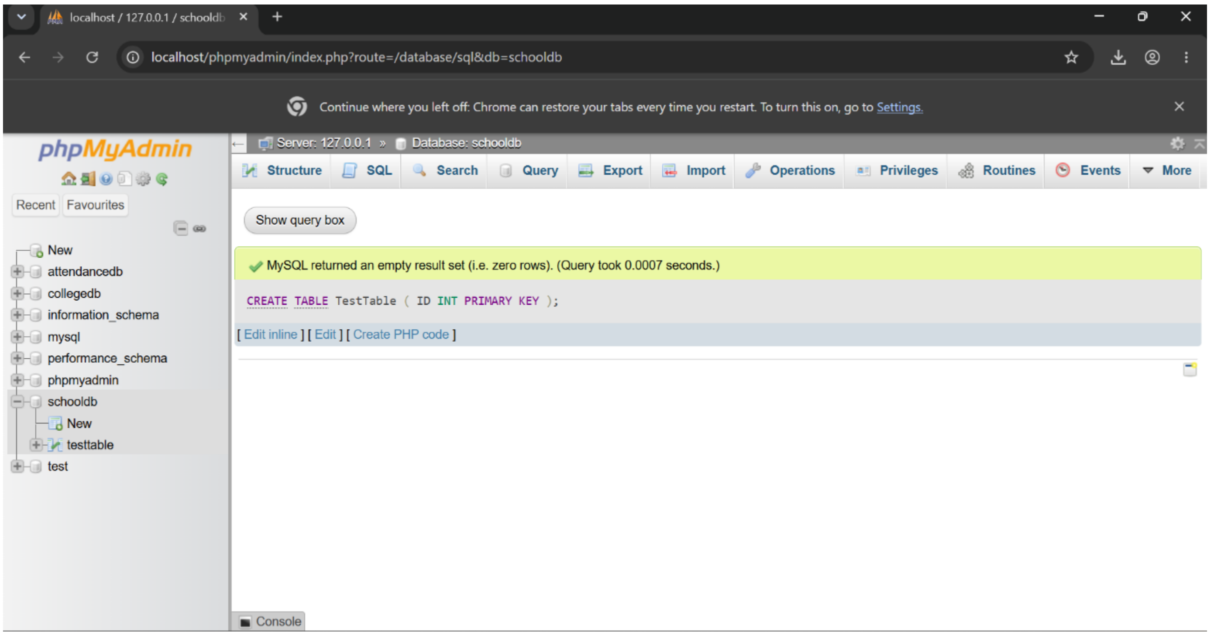
Answer :-SQL Query:

CREATE DATABASE SchoolDB;

USE SchoolDB;

CREATE TABLE TestTable (ID INT PRIMARY KEY);

SHOW TABLES;



Q2) Create a table named Student with the following columns:

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
Age	INT	
ClassName	VARCHAR(20)	
City	VARCHAR(30)	

The program should:

1. Create the table.

2. Define proper data types.

3. Apply primary key on StudentID.

Answer:-

CREATE TABLE Student (

StudentID INT PRIMARY KEY,

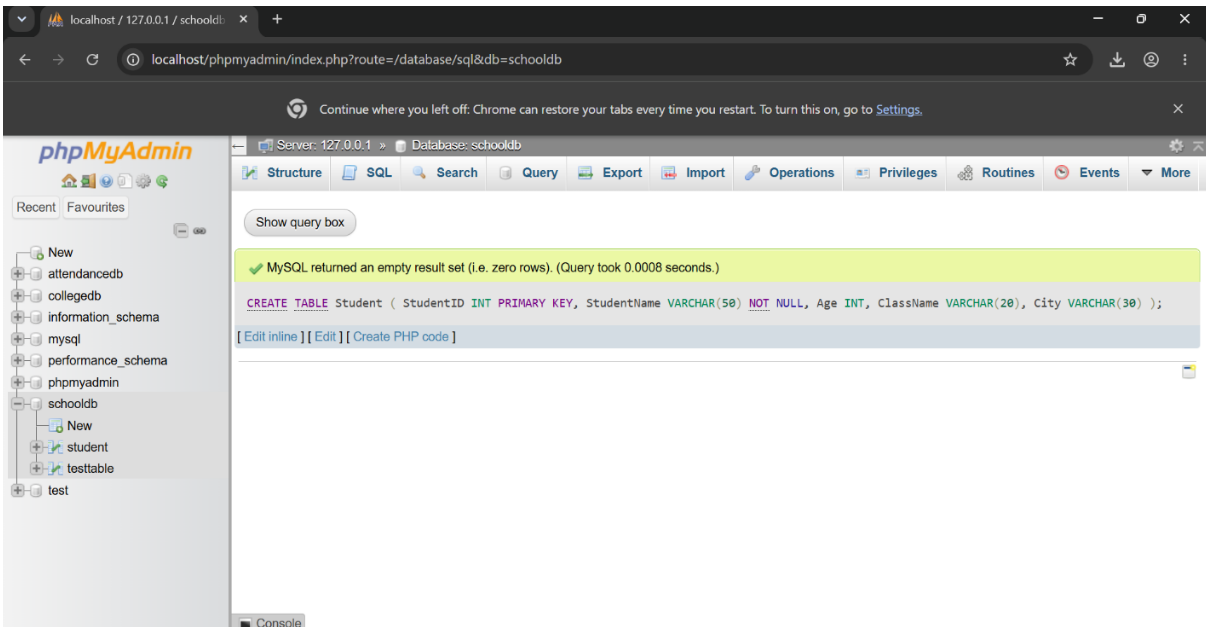
StudentName VARCHAR(50) NOT NULL,

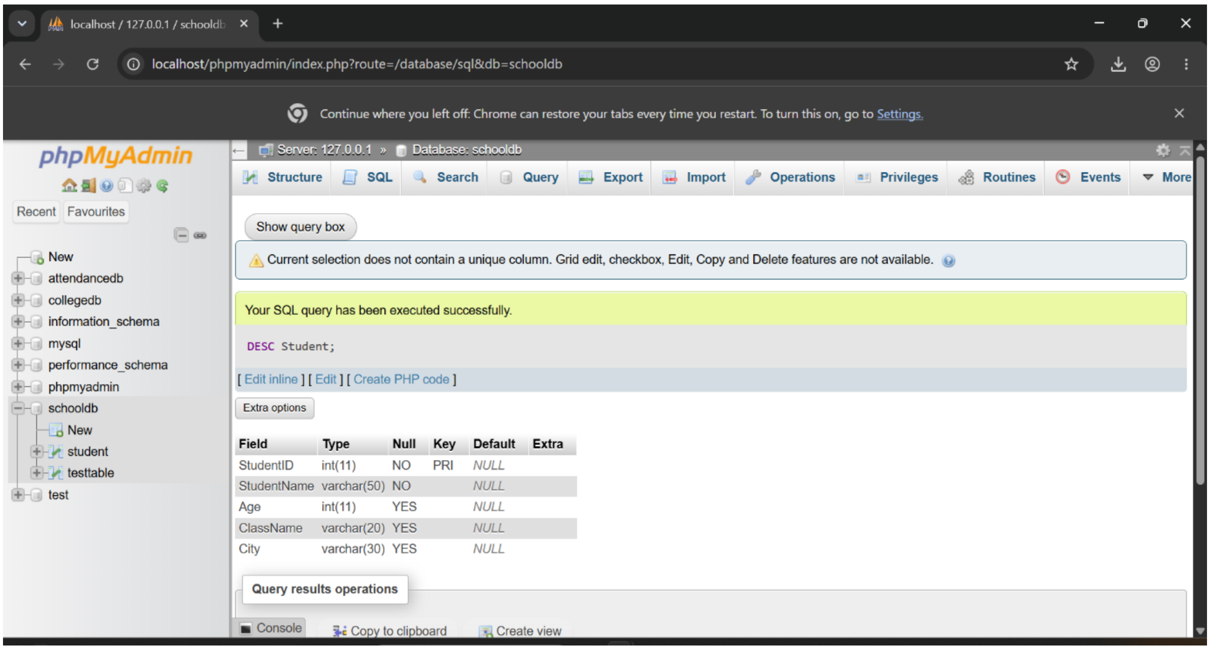
Age INT,

ClassName VARCHAR(20),

City VARCHAR(30));

DESC Student;





Q3) Insert at least 10 records into the Student table. The records should include:

1. Student ID

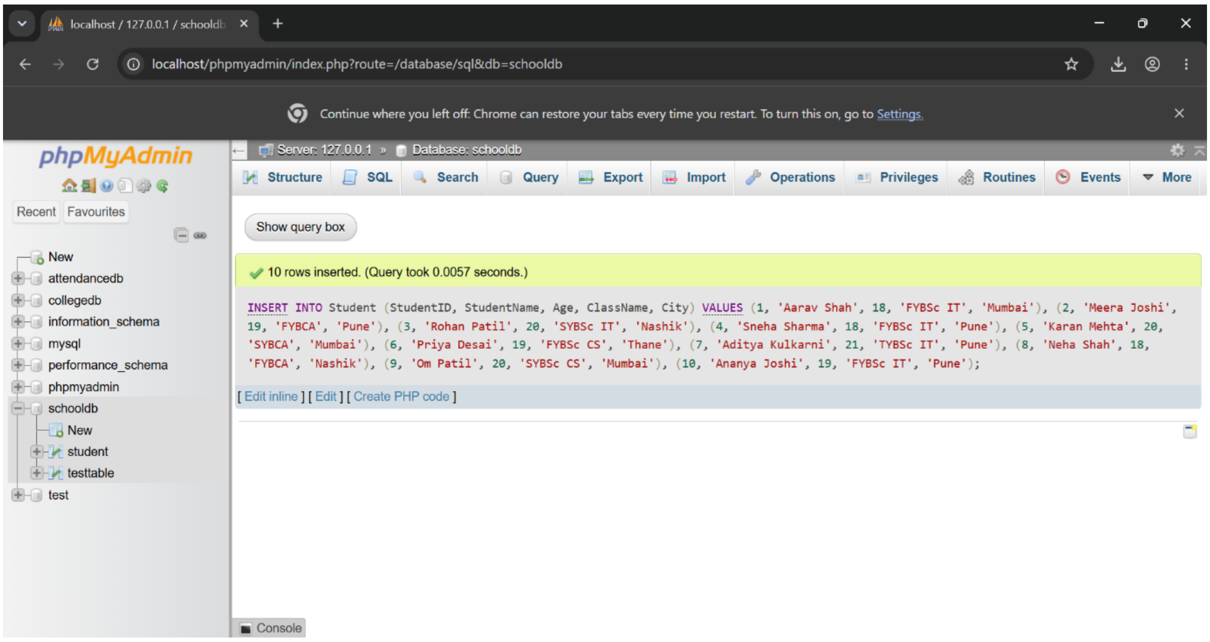
2. Student name

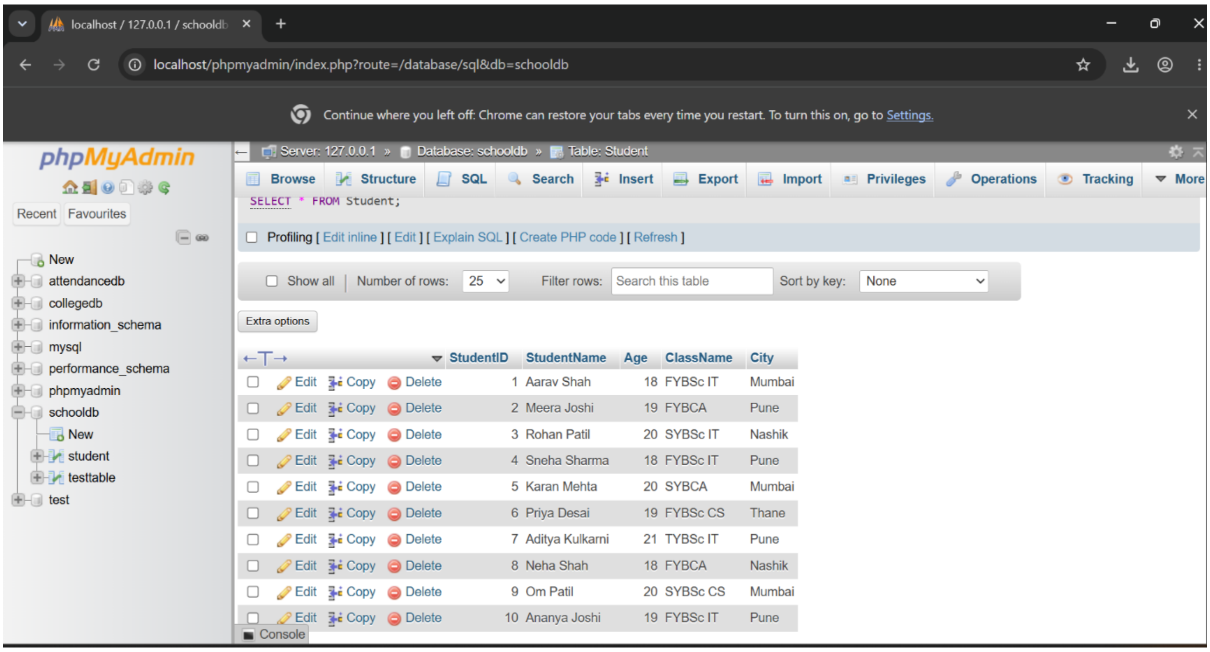
3. Age

4. Class name

5. City After inserting data, display all records using the SELECT command.

Answer:-





Q4) Create a table named Course with the following fields:

Column Name	Data Type
CourseID	INT
CourseName	VARCHAR(50)
Duration	VARCHAR(20)
Fees	DECIMAL(10,2)

The program should:

1. Create the table.

2. Add a primary key on CourseID.

3. Insert at least 5 course records.

4. Display all course records.

Answer:-

CREATE TABLE Course (

  CourseID INT PRIMARY KEY,

  CourseName VARCHAR(50),

  Duration VARCHAR(20),

  Fees DECIMAL(10,2)

);

INSERT INTO Course

(CourseID, CourseName, Duration, Fees)

VALUES

(101, 'BSc Data Science', '3 Years', 45000.00),

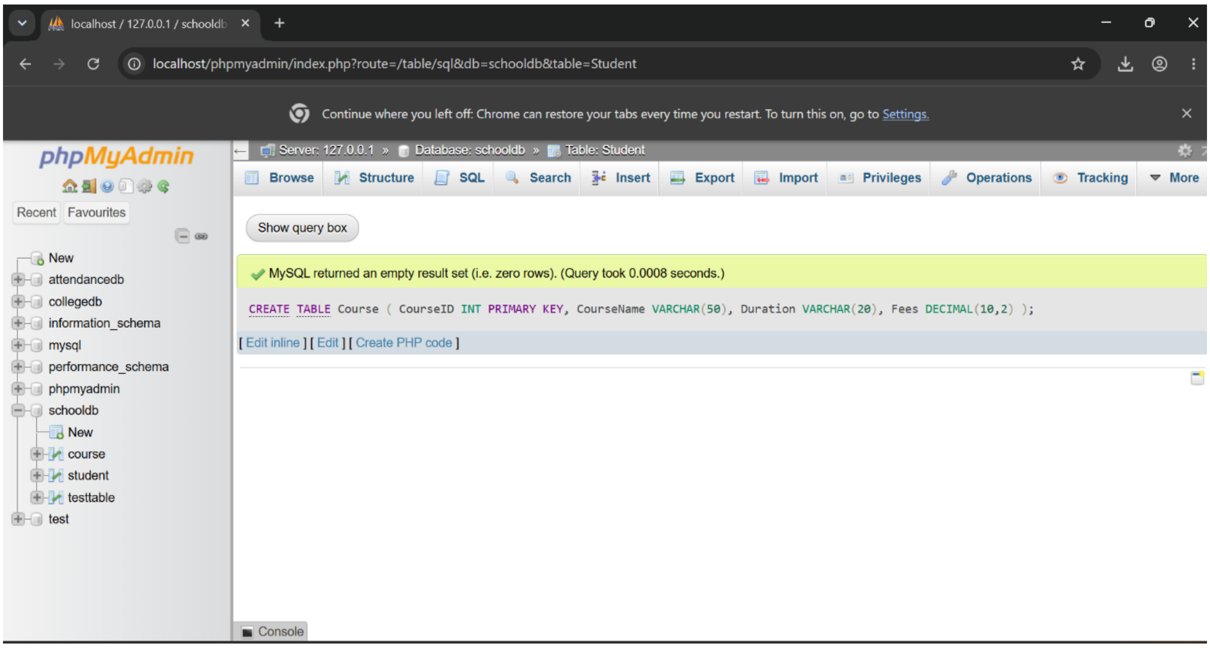
(102, 'BCA', '3 Years', 40000.00),

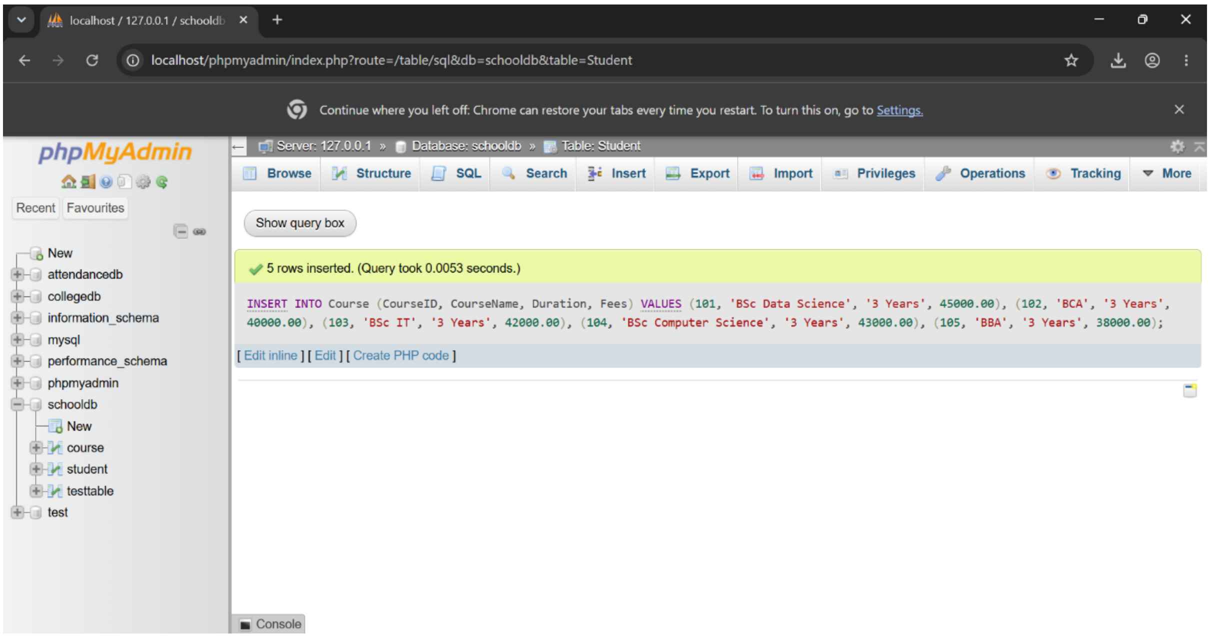
(103, 'BSc IT', '3 Years', 42000.00),

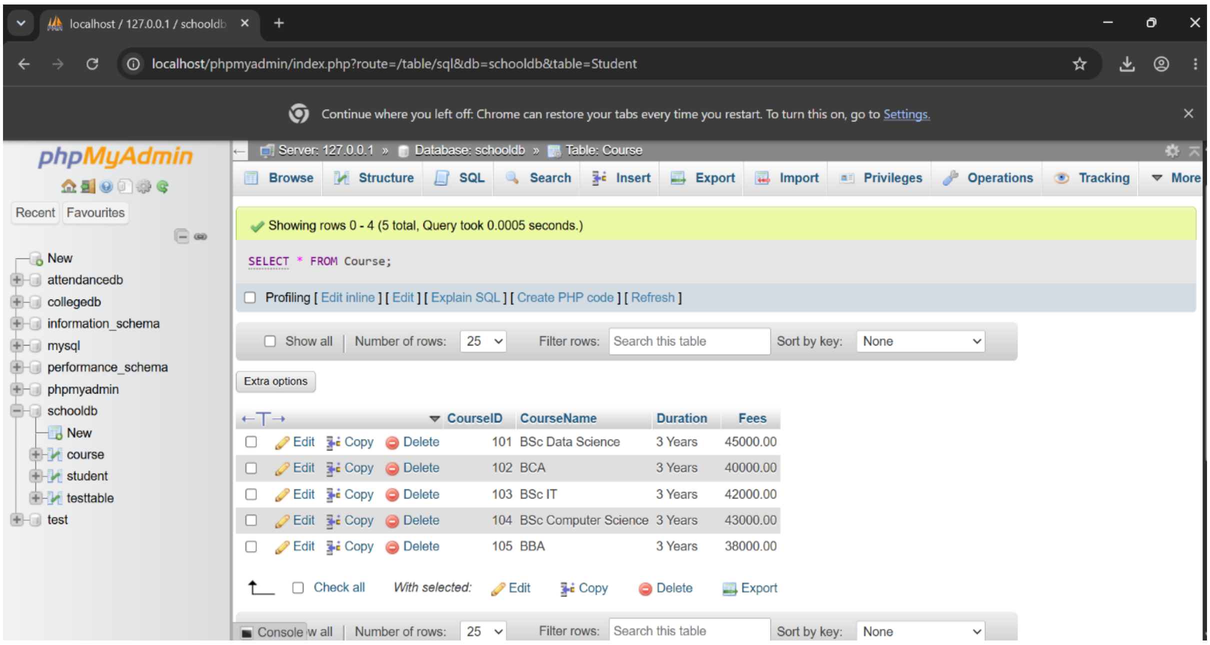
(104, 'BSc Computer Science', '3 Years', 43000.00),

(105, 'BBA', '3 Years', 38000.00);

SELECT * FROM Course;





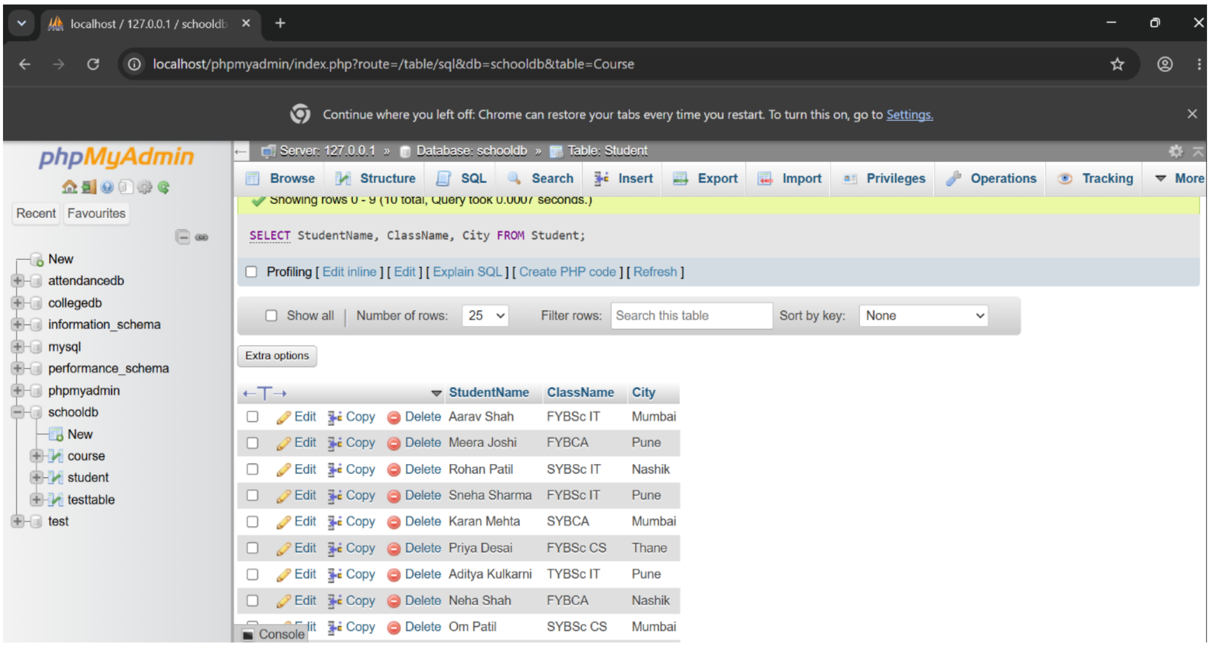


Q5) Write SQL queries to display only selected columns from the Student table.

Display:

1. StudentName
2. ClassName
3. City Expected output should show only these three columns.

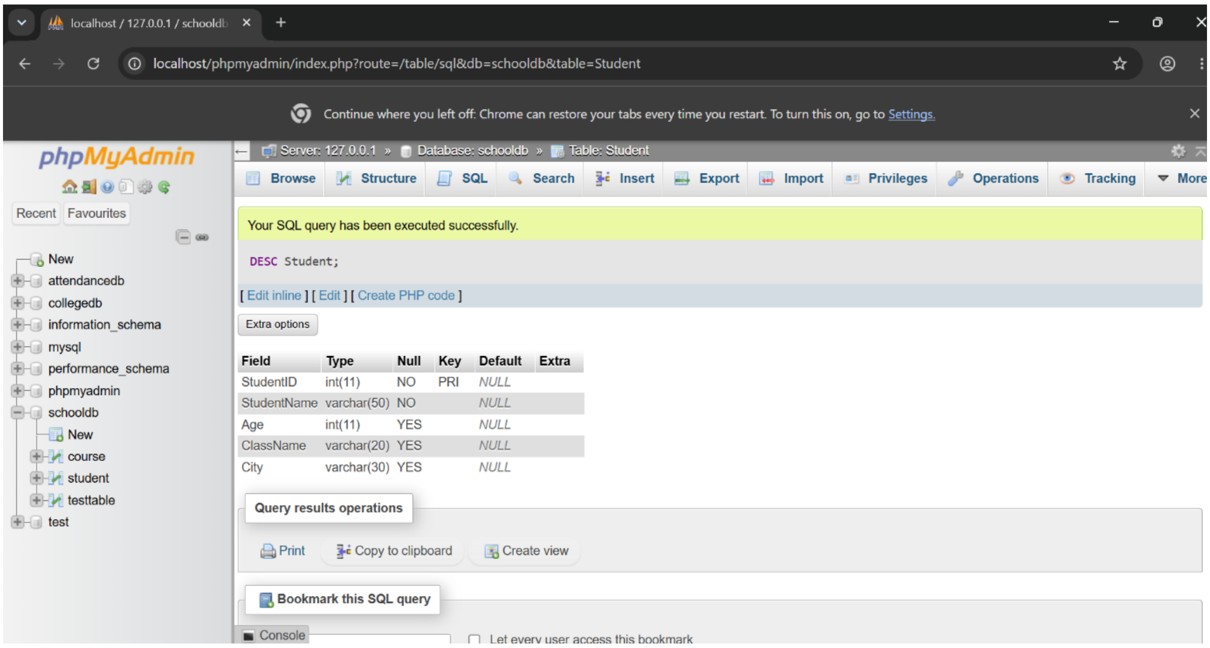
Answer:-



Q6) Write SQL commands to view the structure of the Student table. Use the suitable command according to the database environment:

Database	Command
MySQL	DESC Student;
PostgreSQL	tudent
SQLite	PRAGMA table_info(Student);

Answer:-



Q7) Prepare a CSV file named students.csv and import it into the Student table. The CSV file should contain:

StudentID	StudentName	Age	ClassName	City
1	Aarav Shah	18	FYBSc IT	Mumbai
2	Meera Joshi	19	FYBCA	Pune
3	Rohan Patil	20	SYBSc IT	Nashik

The task should include:

1. Creating a matching table.

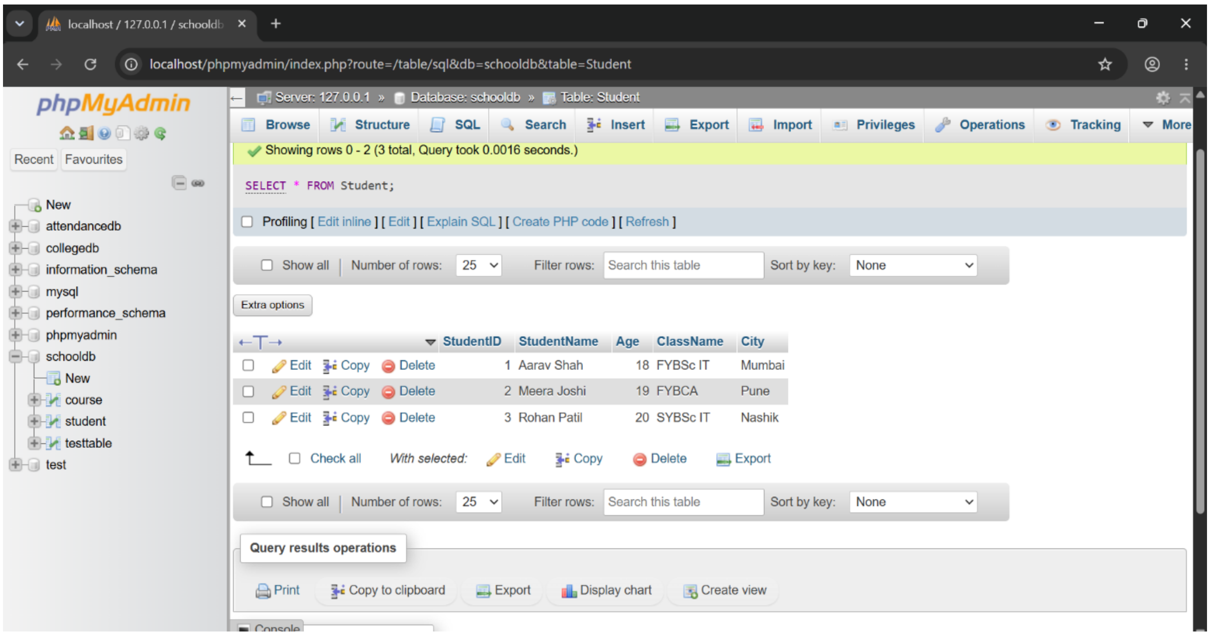
2. Importing the CSV file.

3. Displaying imported records using SELECT.

Answer:-

SQL Query:

SELECT * FROM Student;

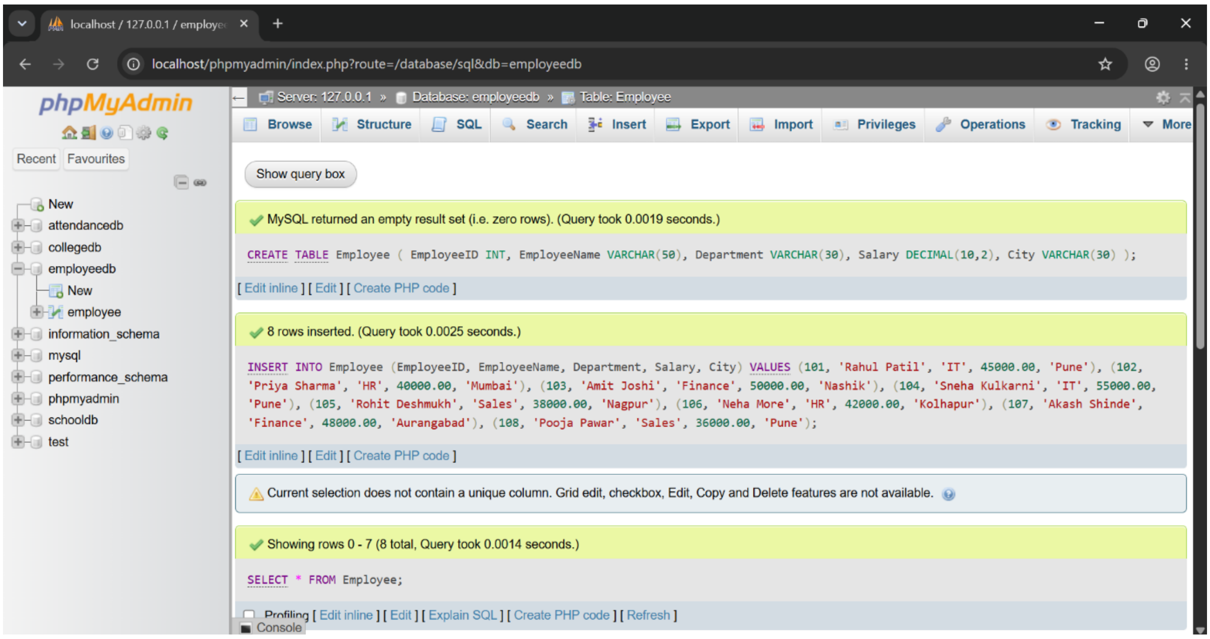


Q8) Create a database named EmployeeDB and a table named Employee. The table should contain

Column Name	Data Type
EmployeeID	INT
EmployeeName	VARCHAR(50)
Department	VARCHAR(30)
Salary	DECIMAL(10,2)
City	VARCHAR(30)

The program should:

1. **Create the database.**
2. **Create the Employee table.**
3. **Insert 8 employee records.**
4. **Display all employee records.**



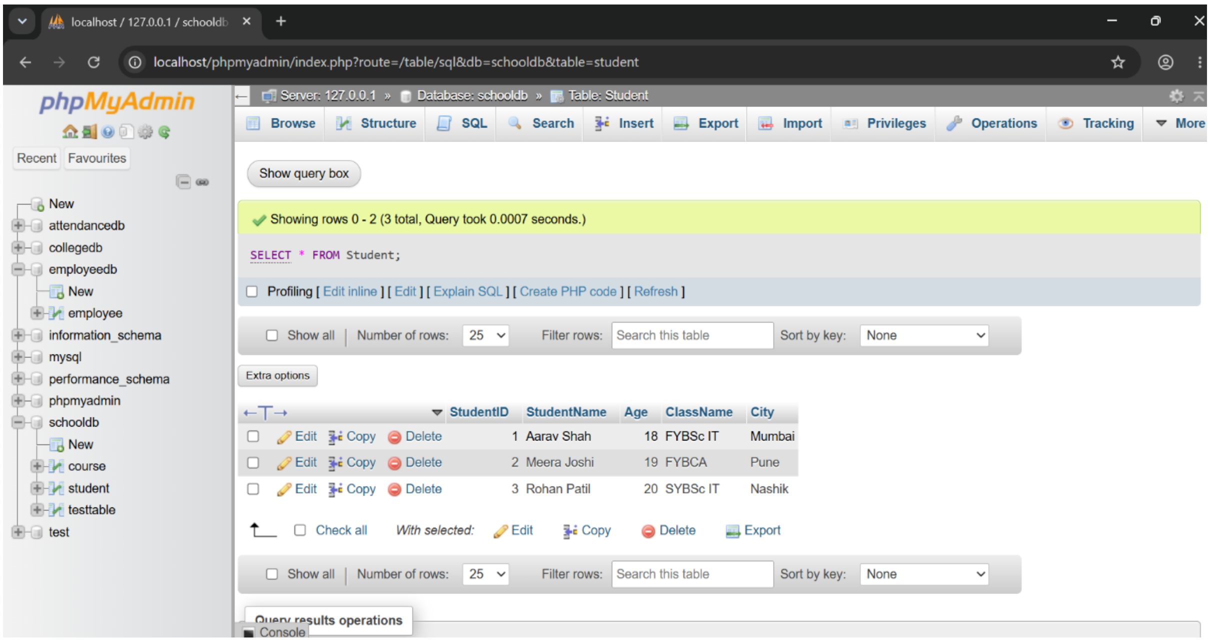
Q9) Write SQL queries to compare the difference between SELECT * and selecting specific columns.

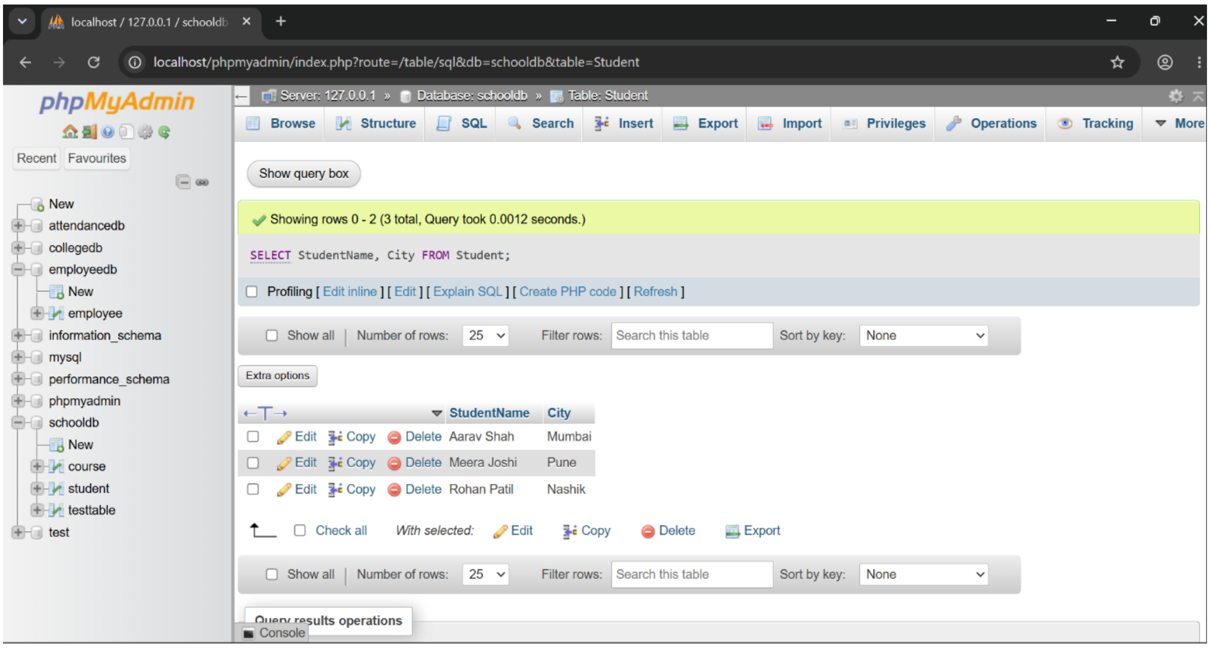
The task should include:

1. Query using SELECT * FROM Student;

2. Query using SELECT StudentName, City FROM Student;

3. Write two lines explaining the difference between both outputs.





1. **SELECT * FROM Student; displays all columns and all records from the Student table.**
2. **SELECT StudentName, City FROM Student; displays only the StudentName and City columns from the Student table.**

Q10) Create a complete SQL script for a mini college database. The script should include:

1. Creating database MiniCollegeDB.

2. Creating tables Student, Course, and Faculty.

3. Adding suitable columns and data types.

4. Inserting at least 5 records in each table.

5. Displaying all records from each table.

6. Viewing the structure of each table.

Answer:-

CREATE DATABASE MiniCollegeDB;

USE MiniCollegeDB;

CREATE TABLE Student (

StudentID INT,

StudentName VARCHAR(50),

Age INT,

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    CourseID INT,
    City VARCHAR(30)
);

CREATE TABLE Course (
    CourseID INT,
    CourseName VARCHAR(50),
    Duration VARCHAR(20),
    Fees DECIMAL(10,2)
);

CREATE TABLE Faculty (
    FacultyID INT,
    FacultyName VARCHAR(50),
    Department VARCHAR(30),
    Salary DECIMAL(10,2)
);

INSERT INTO Student
(StudentID, StudentName, Age, CourseID, City)
VALUES
(1, 'Aarav Shah', 18, 101, 'Mumbai'),
(2, 'Meera Joshi', 19, 102, 'Pune'),
(3, 'Rohan Patil', 20, 103, 'Nashik'),
(4, 'Sneha Kulkarni', 19, 101, 'Pune'),
(5, 'Aditya More', 18, 104, 'Nagpur');

INSERT INTO Course
(CourseID, CourseName, Duration, Fees)
VALUES
(101, 'BSc Data Science', '3 Years', 45000.00),
(102, 'BCA', '3 Years', 40000.00),
(103, 'BSc Computer Science', '3 Years', 42000.00),
(104, 'BBA', '3 Years', 38000.00),
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(105, 'BCom', '3 Years', 35000.00);

INSERT INTO Faculty

(FacultyID, FacultyName, Department, Salary)

VALUES

(201, 'Dr. Amit Joshi', 'Computer Science', 60000.00),

(202, 'Prof. Priya Sharma', 'Data Science', 55000.00),

(203, 'Prof. Rahul Patil', 'Commerce', 50000.00),

(204, 'Dr. Neha More', 'Management', 58000.00),

(205, 'Prof. Sneha Deshmukh', 'Mathematics', 52000.00);

SELECT * FROM Student;

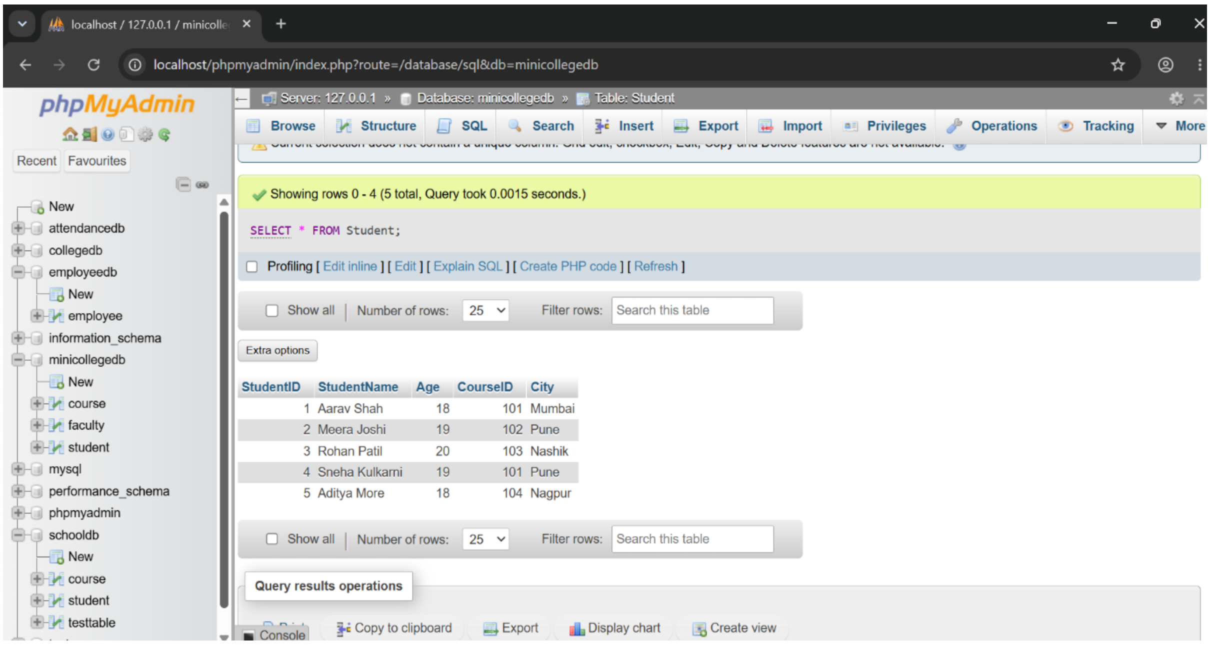
SELECT * FROM Course;

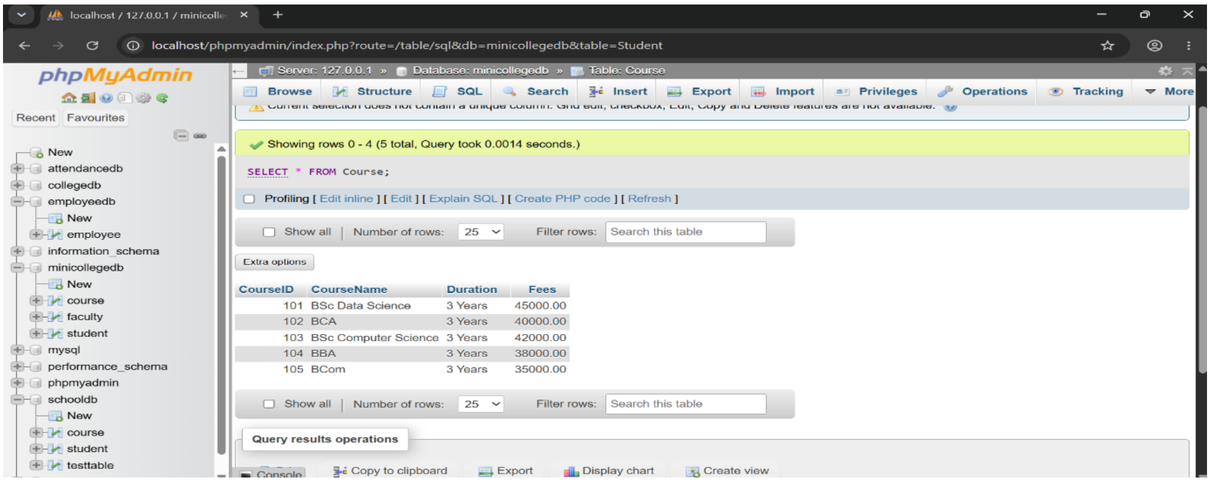
SELECT * FROM Faculty;

DESCRIBE Student;

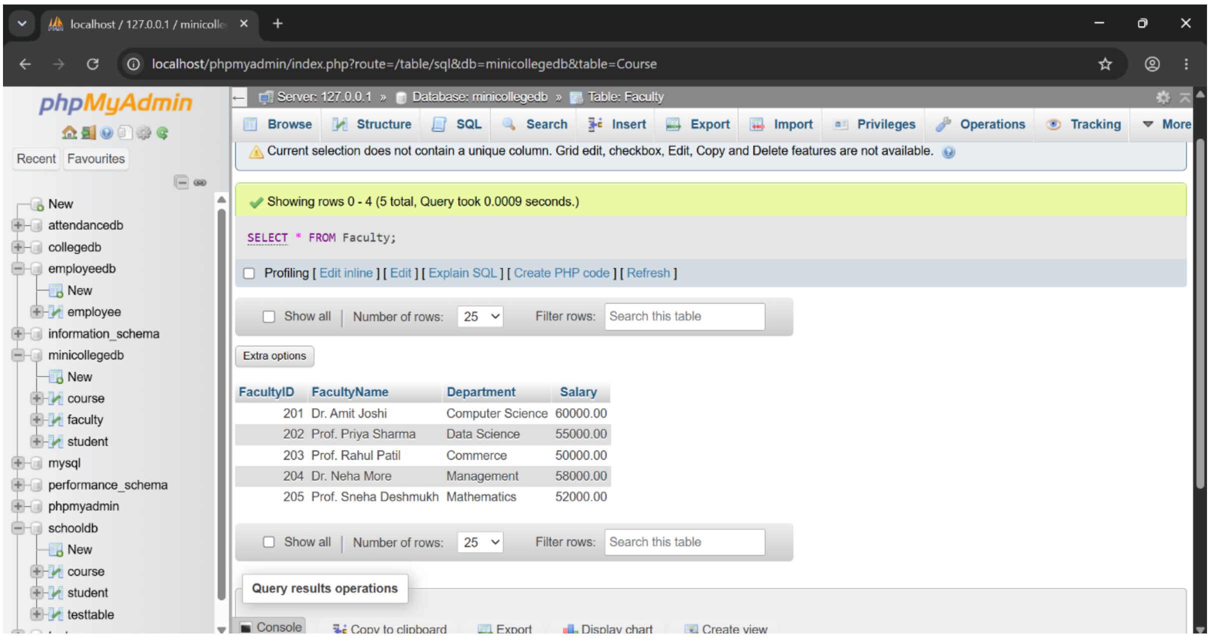
DESCRIBE Course;

DESCRIBE Faculty;

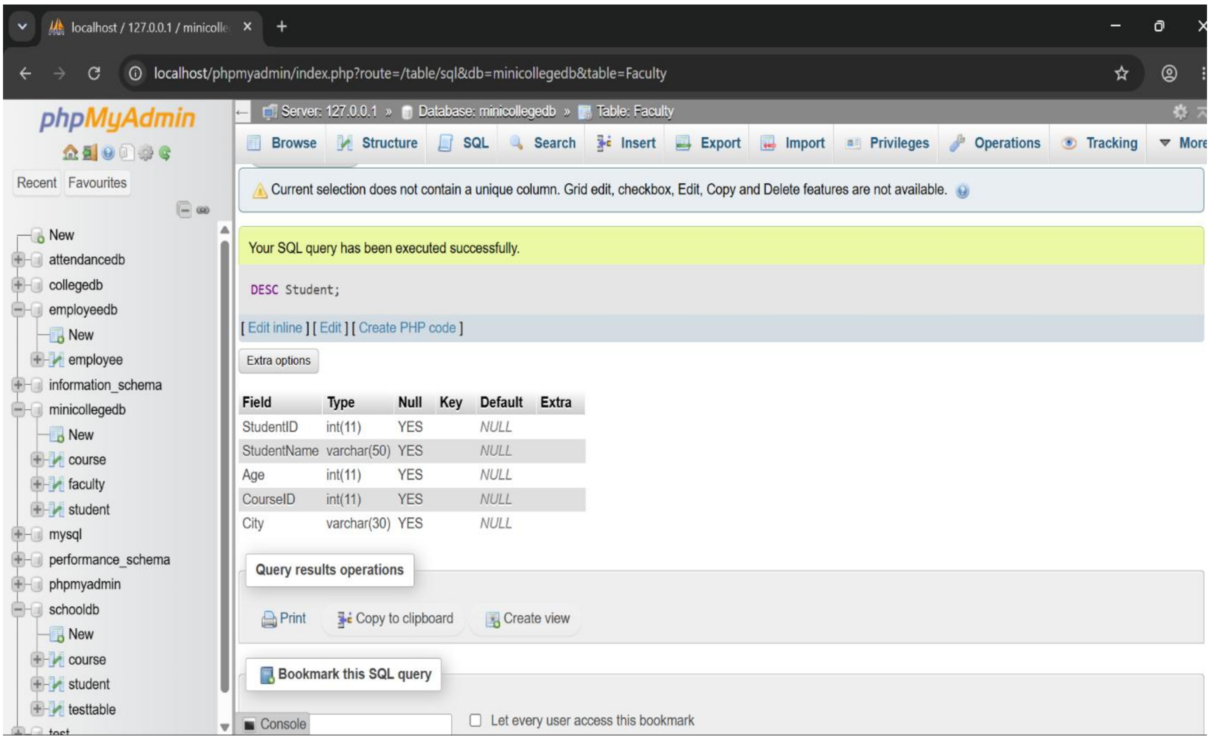




CourseID	CourseName	Duration	Fees
101	BSc Data Science	3 Years	45000.00
102	BCA	3 Years	40000.00
103	BSc Computer Science	3 Years	42000.00
104	BBA	3 Years	38000.00
105	BCom	3 Years	35000.00



FacultyID	FacultyName	Department	Salary
201	Dr. Amit Joshi	Computer Science	60000.00
202	Prof. Priya Sharma	Data Science	55000.00
203	Prof. Rahul Patil	Commerce	50000.00
204	Dr. Neha More	Management	58000.00
205	Prof. Sneha Deshmukh	Mathematics	52000.00


localhost / 127.0.0.1 / minicolle
localhost/phpmyadmin/index.php?route=/table/sql&db=minicollegedb&table=Faculty
phpMyAdmin
Recent Favourites
New
attendancedb
collegedb
employeedb
New
employee
information_schema
minicollegedb
New
course
faculty
student
mysql
performance_schema
phpmyadmin
schooldb
New
course
student
testtable
Server: 127.0.0.1 » Database: minicollegedb » Table: Faculty
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More
Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.
Your SQL query has been executed successfully.
DESC Student;
[Edit inline] [Edit] [Create PHP code]
Extra options
Field Type Null Key Default Extra
StudentID int(11) YES NULL
StudentName varchar(50) YES NULL
Age int(11) YES NULL
CourseID int(11) YES NULL
City varchar(30) YES NULL
Query results operations
Print Copy to clipboard Create view
Bookmark this SQL query
Console
Let every user access this bookmark